

Convolution

1. Convolutional layer can be thought of as an analog of linear layer for 1D vectors, although there are some subtleties like weight sharing and sliding window approach for memory to be tractable in processing images.
2. Convolution formula $W_{out} = \frac{W_{in} + 2P + K}{stride} + 1$.
3. Tip: By padding the input image with $K//2$ the output will be of the same width and height with the input. example:
 1. padded image size = $10 + 3//2 + 3//2 = 12$
 2. convolve with kernel of size $K = 3$.
 3. get original size back: $W_{out} = \frac{12+0-3}{1} + 1 = 9 + 1 = 10$
4. Tip: By padding the output with $K - 1$, and convolving it with the kernel will return an feature map with the same size as the original input. This property is used in conv backprop, example:
 1. Original input size: $W_{in} = 10$
 2. $kernel_size = 3$
 3. $W_{out} = \frac{10+0-3}{1} + 1 = 8$
 4. now pad output with $K - 1$,
 5. $W_{out_padded} = 8 + 2 + 2 = 12$
 6. convolve resulting output with kernel size of K :
 7. $\frac{12+0-3}{1} + 1 = 10$.
 8. Since in backpropagation, $dLdx = dLdz \otimes flipped(W)$, where W is the filter.
5. How big is our receptive field?
 1. $r_0 = \sum_{l=1}^L ((k_l - 1) \prod_{i=1}^{l-1} s_i) + 1$
 1. r_0 is receptive field size
 2. k_l is kernel size at layer l .
 3. $\sum_{i=1}^L$ sum over network layers.
 4. $\prod_{i=1}^{l-1}$ is the product of strides up to layer l .